

~~Multiple 2 P~~

Multiple 2 Partial Correlation.

Multiple correlation is used to study the effect of a group of variable on a single variable.

In multi-variate distribution ~~we are interested in the relationship~~ or in bi-variate distribution we are interested to study the relationship between two variables only. There are two ways.

① Consider those two variables only ~~and~~ in which other members have specified values. JUN.SUN.10

② or eliminate the effect of other variables on two variables.

The below method is partial ~~correlation~~ regression problem. The correlation is called partial correlation.

Appointments

The problem of multivariate distribution is three type.

① To estimate the value of one variable from those of several others. i.e. multiple regression.

Phones to make

② To measure the combined effect of a group of variables upon another variable. This is the problem of multiple correlation.

③ To examine the influence of one variable upon another, after eliminating the effects of all other variables. This is the problem of partial ~~regression~~ correlation.

Remarks

Appointments

Coefficient of multiple correlation:

In bivariate ~~data~~ distribution in which each of the variables

X_1, X_2 and X_3 has n no. of obs. The multiple correlation coefficient of X_1 on X_2 & X_3 is denoted as $R_{1.23}$. This is the simple correlation coefficient of X_1 ~~on X_2 & X_3~~ and the joint effect of X_2 & X_3 on X_1 .

Phones to make

Remarks

$$\therefore R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{23}r_{13}}{1 - r_{23}^2}}$$

$$0 \leq R_{1.23} \leq 1$$

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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 •

multiple correlation interms of
total & partial correlation,
Thursday $1 - R_{1.23}^2 = (1 - r_{12}^2)(1 - r_{13.2}^2)$ 2001

Appointments

$$r_{23.1} = \frac{r_{23} - r_{21}r_{31}}{\sqrt{(1-r_{21}^2)(1-r_{31}^2)}}$$

~~$r_{12.3}$~~ $-1 \leq r_{12.3} \leq 1$

Phones to make

Ex Given $r_{12} = 0.65$, $r_{13} = 0.60$

$r_{23} = 0$

find $R_{1.23}$, $r_{12.3}$,
 $r_{13.2}$, $r_{23.1}$

Ex. If $r_{12} = 0.8$, $r_{13} = -0.4$,

and $r_{23} = -0.56$ find the
values of $r_{12.3}$, $r_{13.2}$, and $r_{23.1}$

also compute $R_{1.23}$, $R_{2.13}$ & $R_{3.12}$.

Remarks